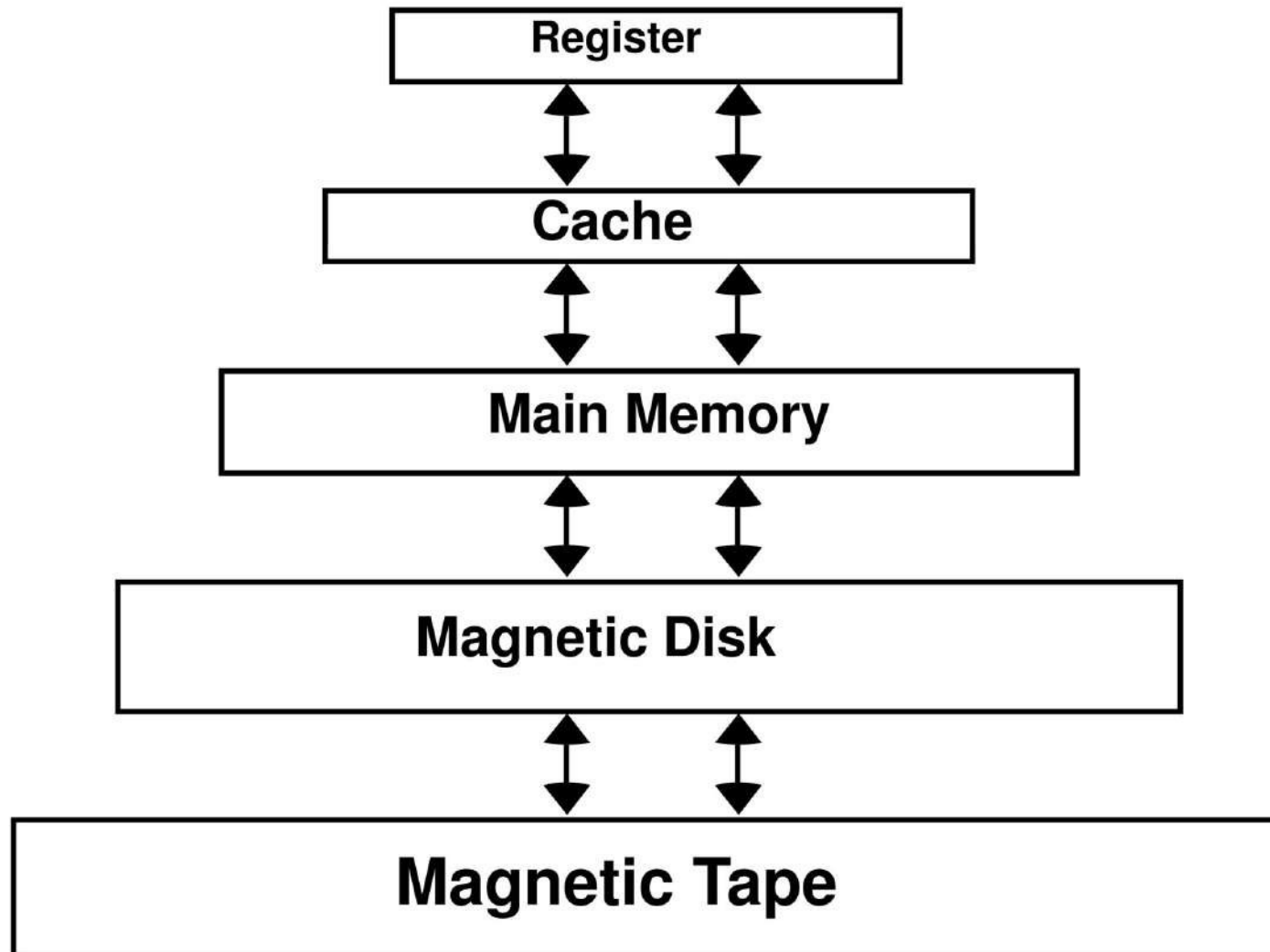


MEMORY SYSTEMS

Prof. Shahenda Sarhan

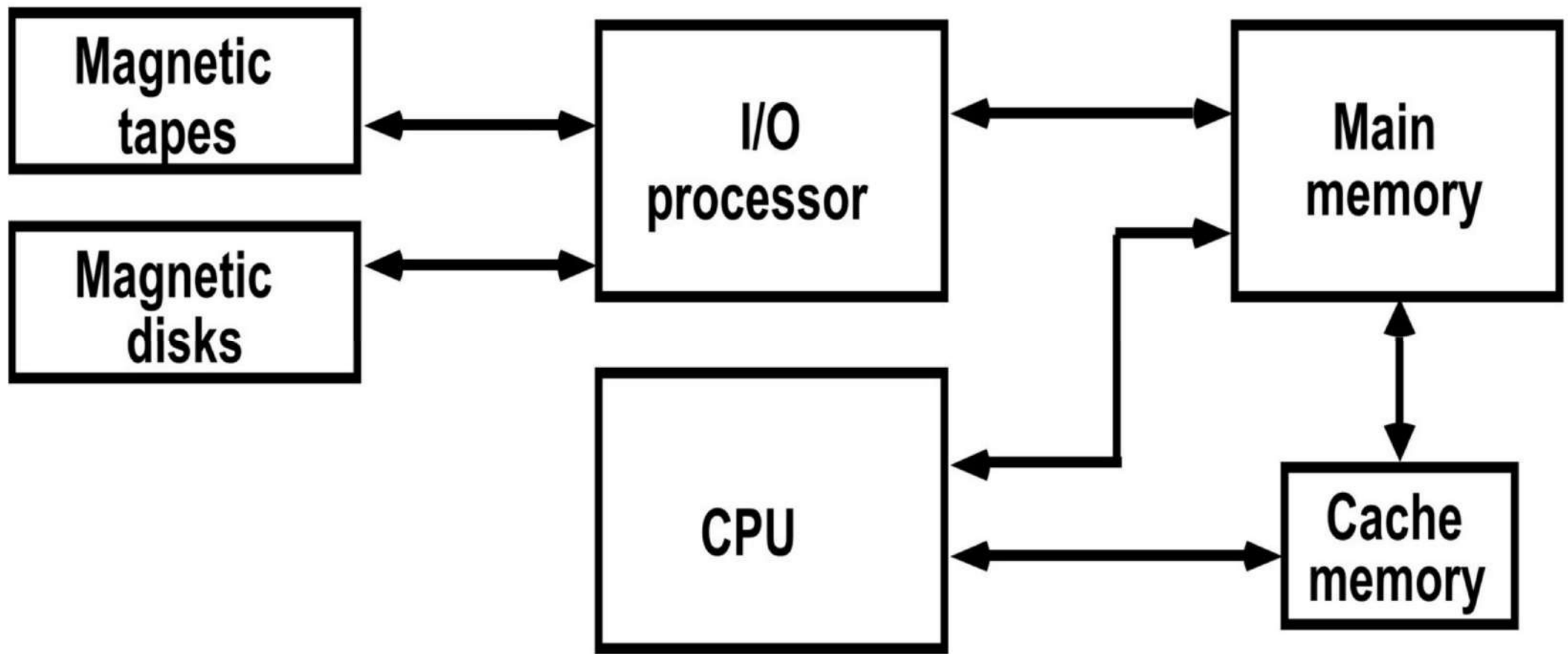
Prof @ computer Science dept.

Memory Hierarchy



Memory Hierarchy

- Access Type: the action that physically takes place, during a read or write operation.
- Capacity of a memory level is usually measured in bytes.
- Cycle Time : time elapsed from the start of a read operation to the start of a subsequent read.
- Latency : time interval between the request for information and the access to the first bit of that information.
- Bandwidth : a measure of the number of bits per second that can be accessed.
- Cost of A Memory Level : specified as dollars per megabytes. Where greater performance is realized, at a greater cost as we move up through the hierarchy.



- The small, fast cache memories act as staging areas for a subset of the data and instructions stored in the relatively slow main memory.
- The main memory acts as staging area for data stored on large, slow disks, which in turn often serve as staging areas for data stored on the disks or tapes of other machines connected by networks.

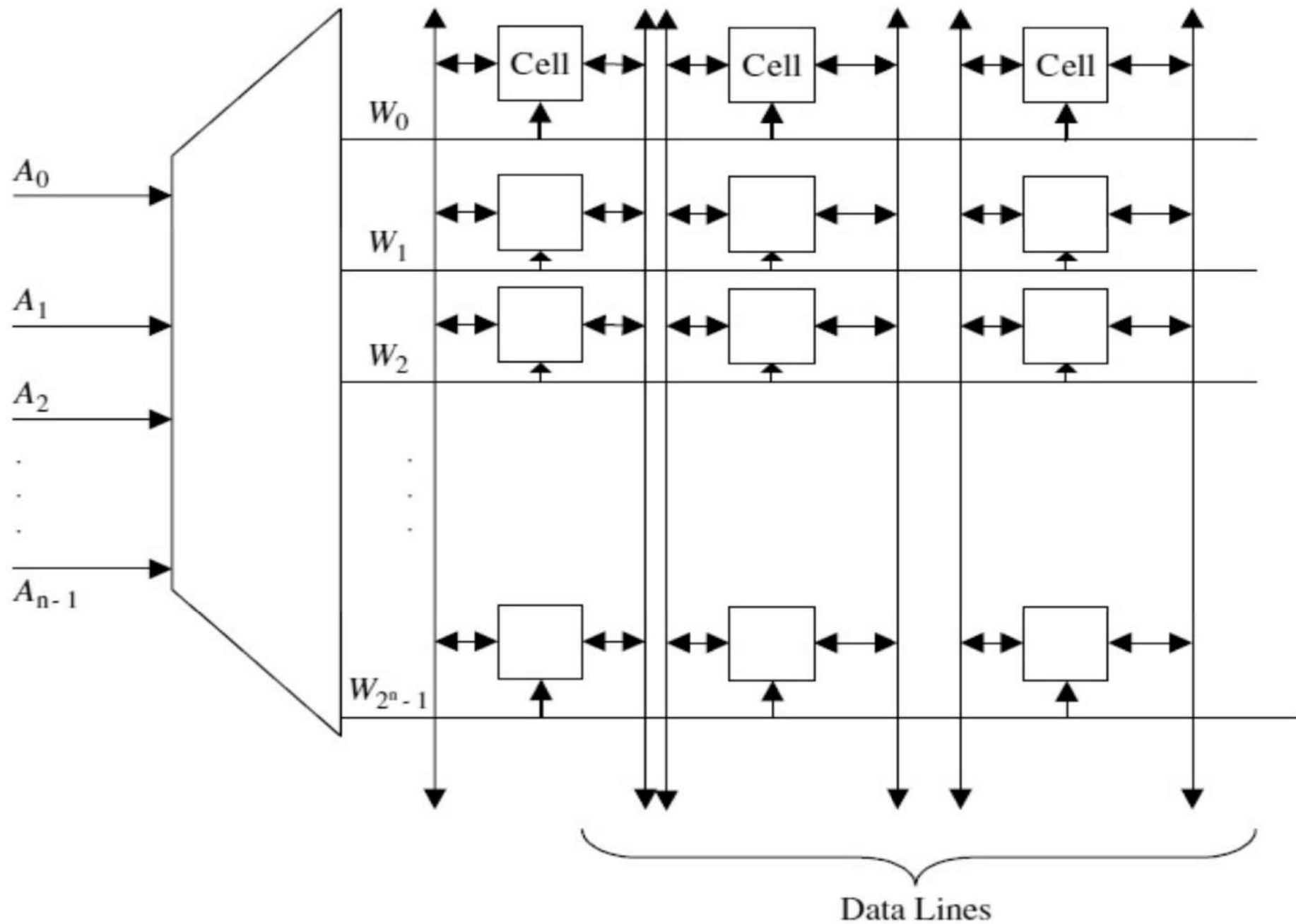
Main Memory

- Memory :is organized in a matrix of rows and columns.
- Each word of memory is uniquely identified by the combined row and column values - known as the ADDRESS.
- Address lines $A_{n-1}, A_{n-2} \dots A_1, A_0$ are used as inputs to the address decoder in order to generate the word select lines $W_{2^n-1}, \dots W_1, W_0$.
- A given word select line is common to all memory cells in the same row.

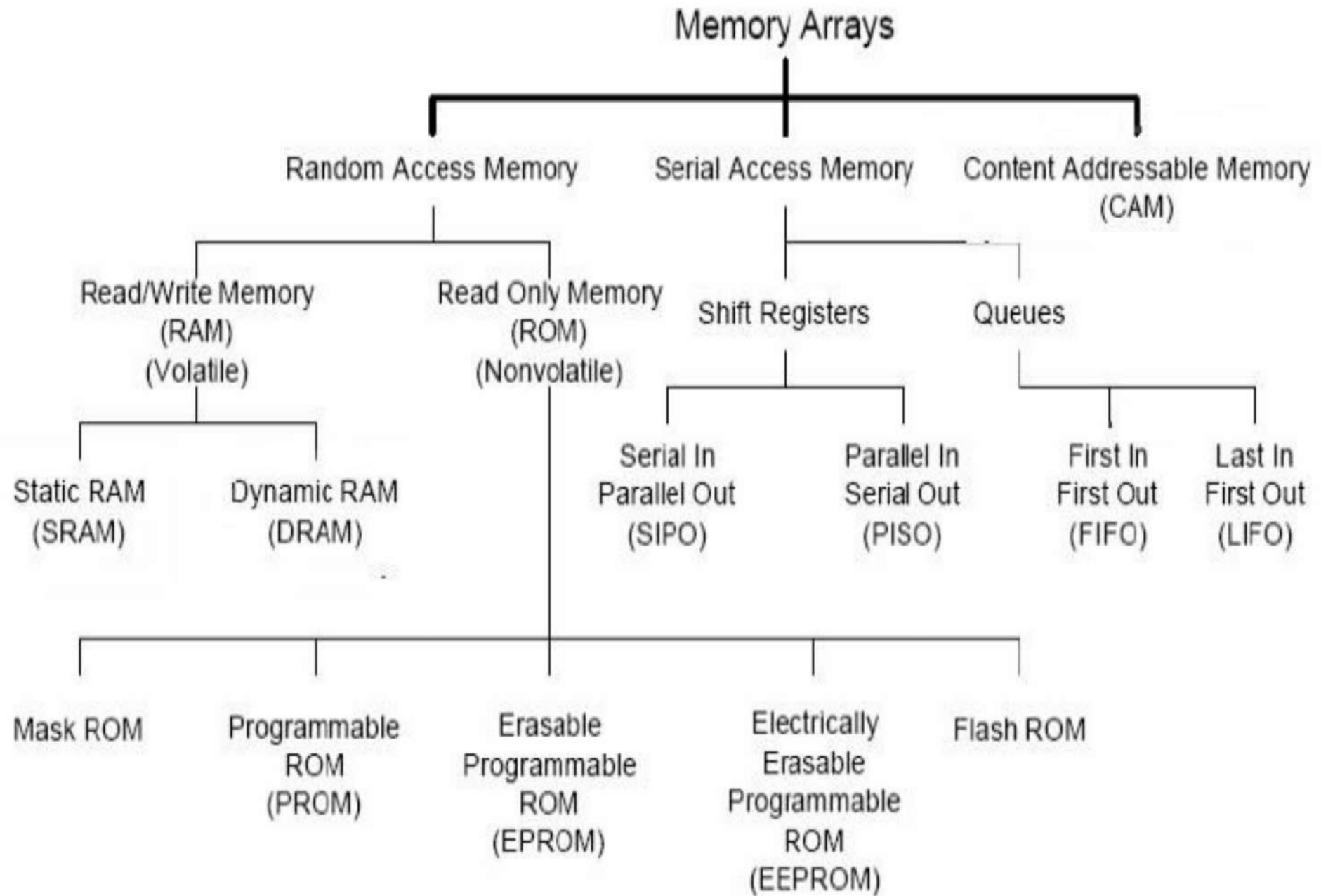
Main Memory

- the Address Decoder activates only one word select line while deactivating the remaining lines.
- A word select line is used to enable all cells in a row for read or write.
- Data lines are used to input or output the contents of cells.
- Each memory cell is connected to two data lines.
- A given data line is common to all cells in a given column.

Main Memory



Main Memory types



Random-access Memory

- The random-access memory could be classified into :
 - Read-write memory : commonly known as RAM
 - Read-only memory : known as ROM

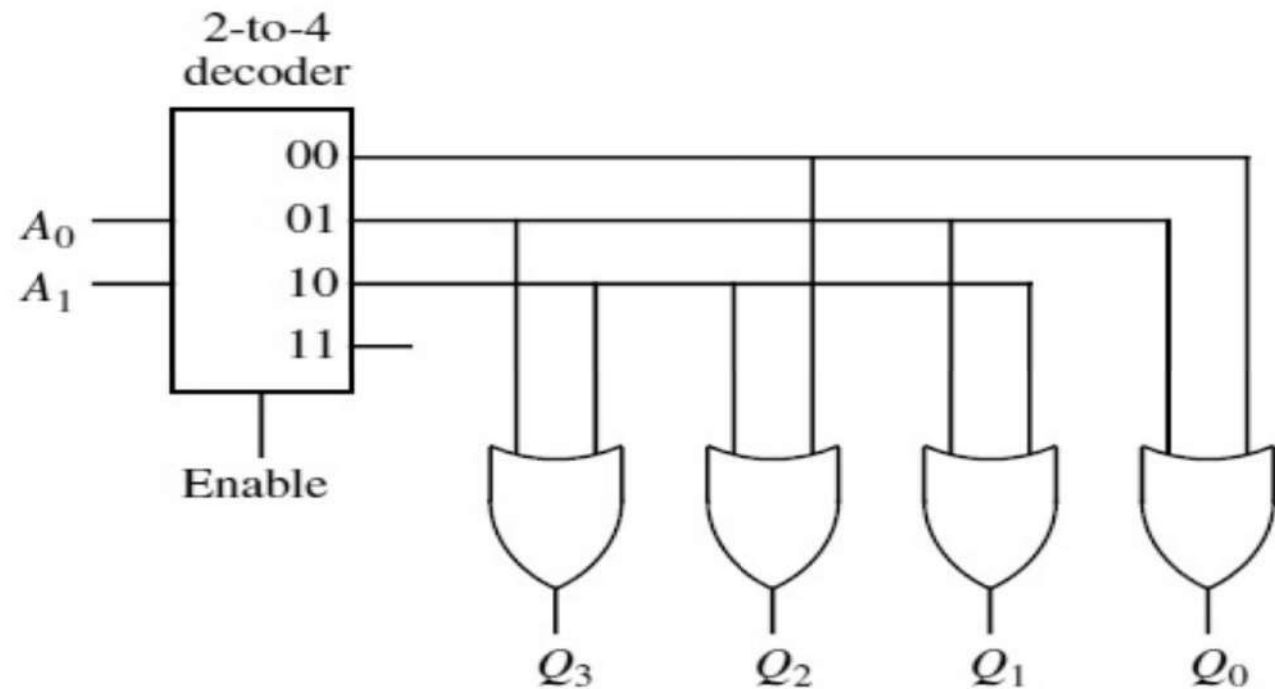
Random Access Memory [Read-Write]

- Access Time is constant for all RAM locations.
- In contrast to register memory, RAM is used to hold both program code (instructions) and data (numbers, strings etc).
- Programs are loaded into RAM from a disk prior to execution by the CPU.
- Locations in RAM are identified by an addressing scheme.
- The contents of RAM are lost if the power is turned off (volatile).
- Types:
 - Static random access memory (SRAM)
 - Dynamic random access memory (DRAM)

SRAM	DRAM
Expensive	cheap
Need no refresh	Need to be refreshed
Shorter access time	Longer access time
Smaller amount of data	Larger amount of data
	SDRAM ,DDR SDRAM

Read-Only Memory [ROM]

- ROM is a form of semi-conductor that can be written to once.
- Store boot or start-up program that a computer executes when powered on.
- Nonvolatile.

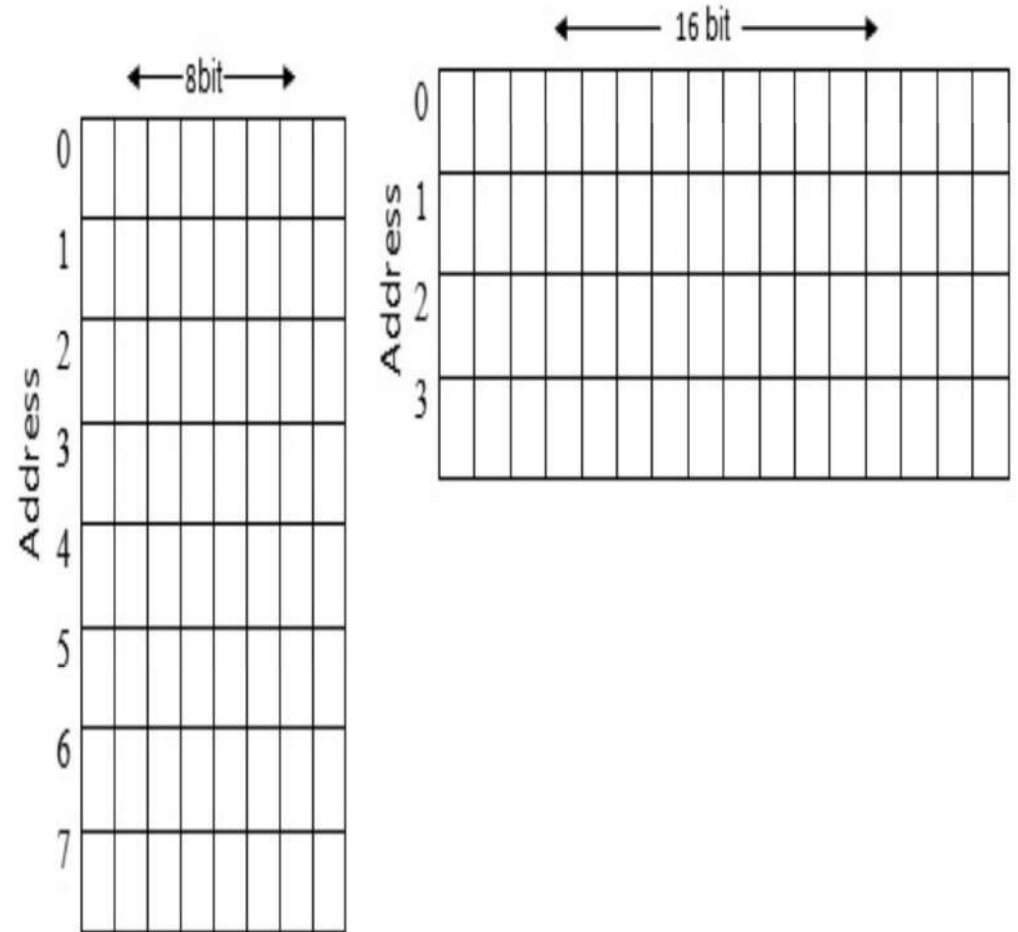


Read-Only Memory Types

- PROM - Programmable Read Only Memory: can only be programmed once.
- EPROM - Erasable Programmable Read Only Memory:
 - rewritten many times and have to be removed to be rewritten
 - Erasing an EPROM requires a special tool that emits a certain frequency of ultraviolet light..
 - An EPROM eraser is not selective , it will erase the entire EPROM.
- EEPROM - Electrically Erasable Programmable Read Only Memory:
 - The chip does not have to be removed to be rewritten.
 - The entire chip does not have to be completely erased to change a specific portion of it.

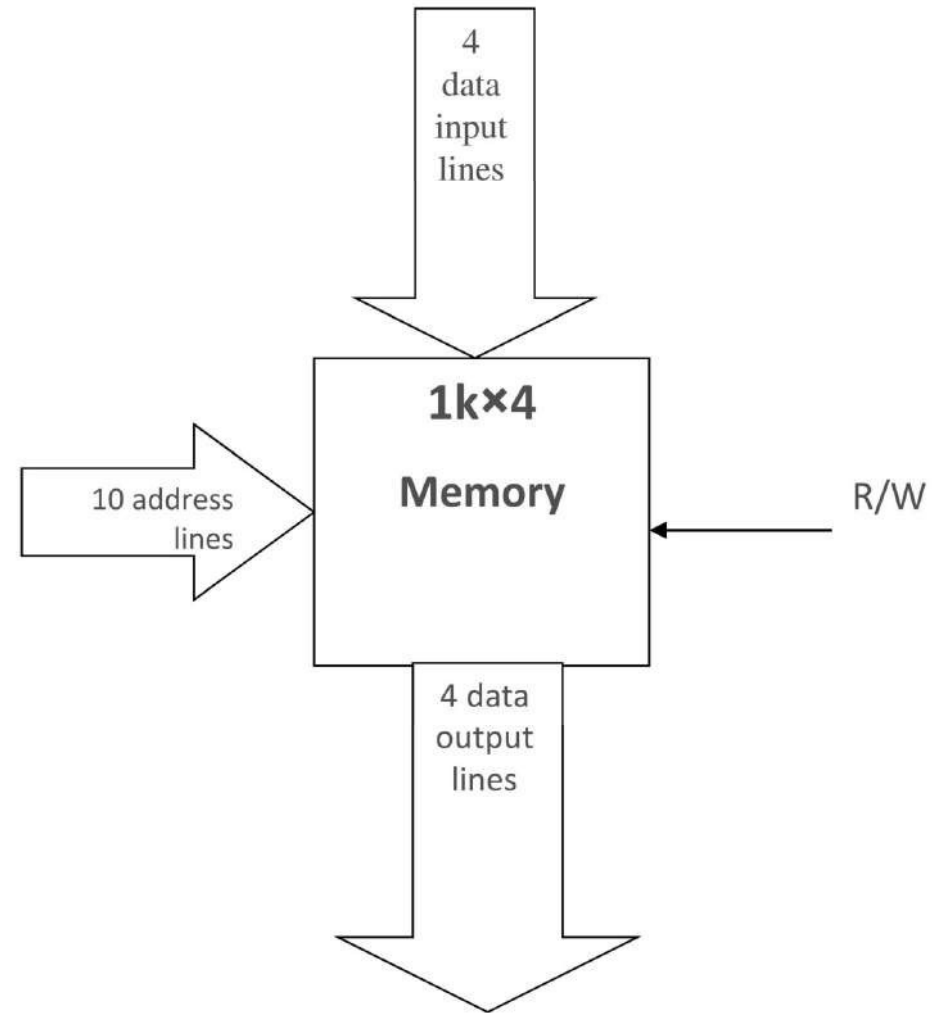
Memory Organization

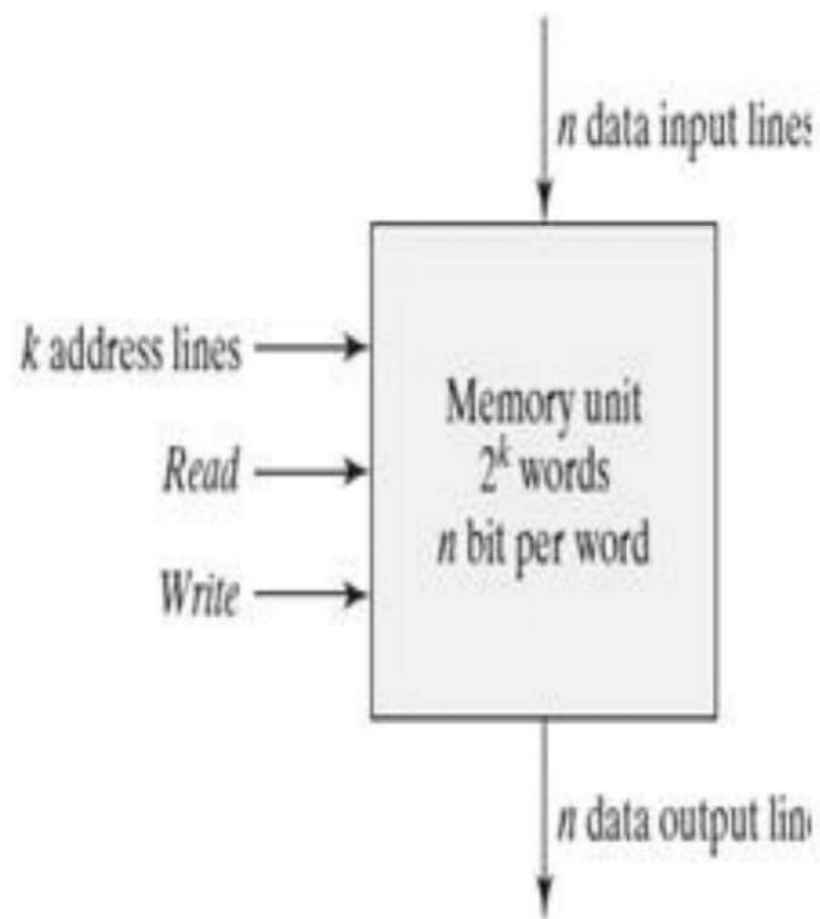
- Memory can be considered to be organized as a matrix of bits
 - Ex: 64 bits memory could be 8×8 or 16×4 or 4×16 .
- Each row represents a memory location
 - the number of bits in row is the word size of the architecture
 - It can be a word multiple for example two words, four words, etc...



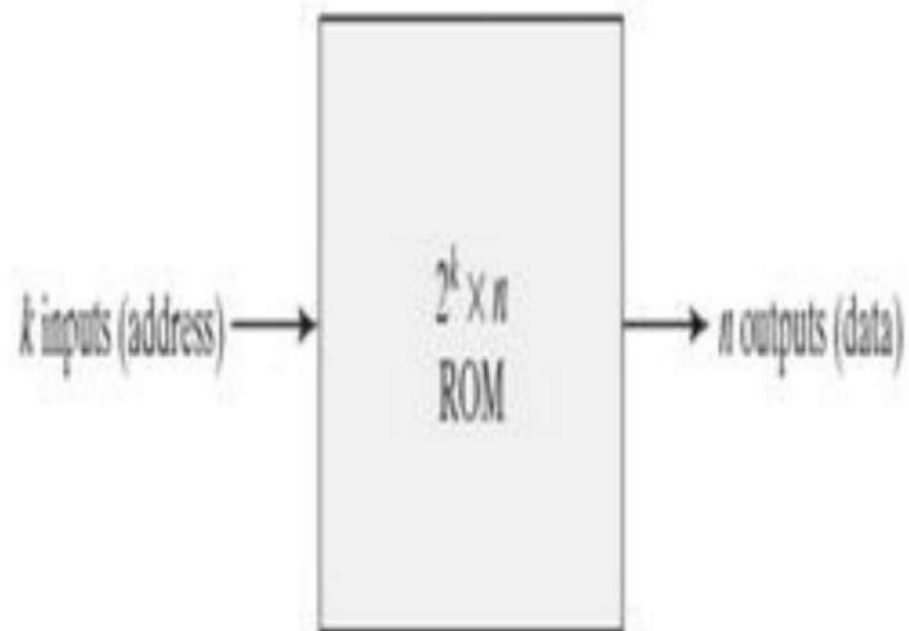
Memory Organization

- Internal organization of the memory array should satisfy the efficient utilization of the memory chip.
 - Ex: a $1K \times 4$ [$2^{10} \times 4$] memory chip may be organized as
 - 2^{10} rows of cells, each consisting of four cells.
 - 10 pins for the address and four pins for the data.





RAM



ROM

Memory Organization

$$\text{No of chips per memory module} = \frac{\text{size of given memory}}{\text{size of memory chip}}$$

- Ex: the design of a 4M bytes [4M×8] main memory subsystem using 1M bit chip.

$$\frac{4\text{M} \times 8}{1\text{M} \times 1} = 32 \text{ Chips}$$

